

Lecture 16 - Economics and Business Administration

Cost Measurment Systems

Indirect Cost Allocation Strategies in CMS

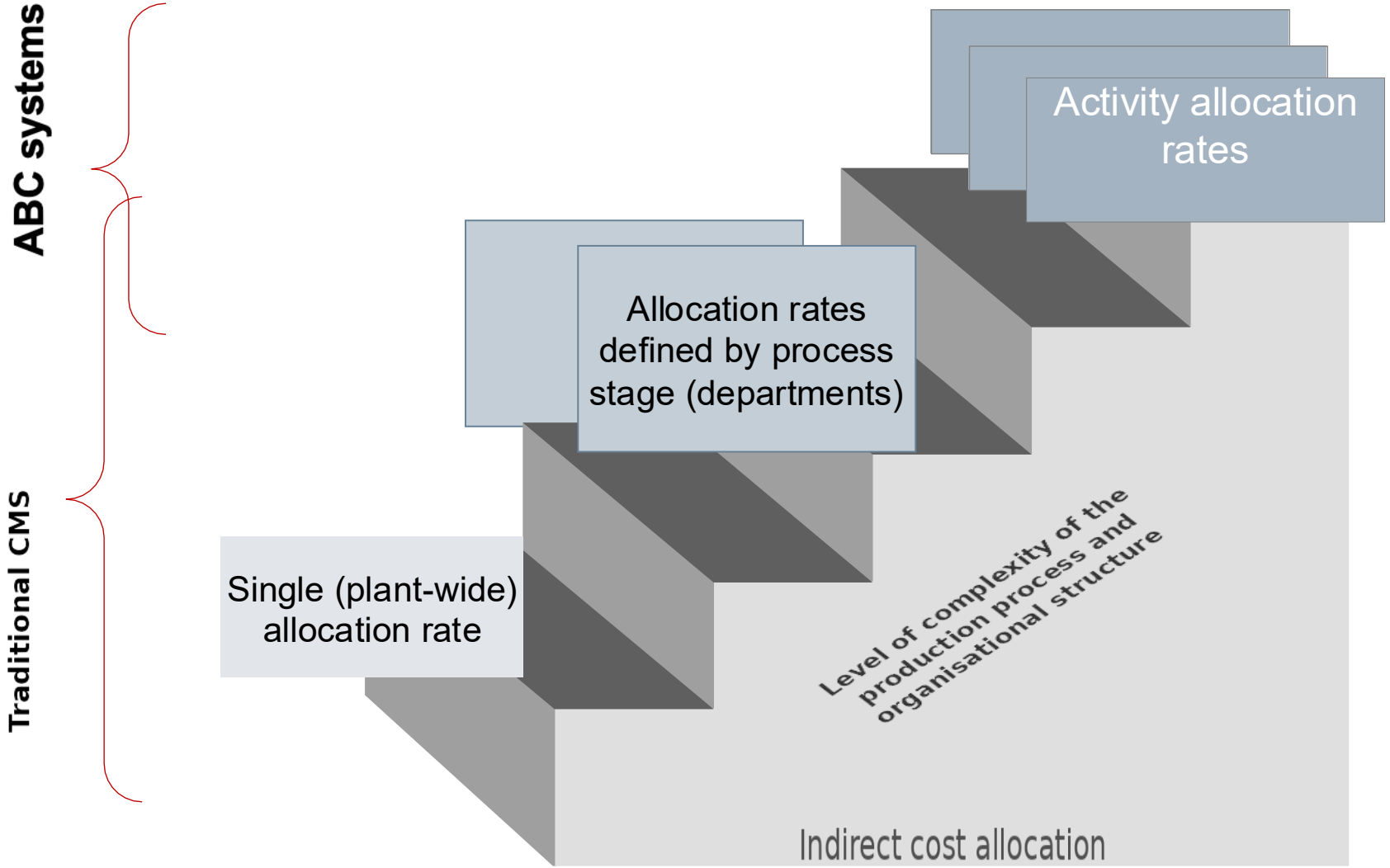
Traditional CMS

- Traditional CMS define the methods for determining the **full manufacturing cost** of a product.
- They therefore focus **on production costs**, treating non-production costs with often simplified methodologies.
- In **very simple businesses** (few cost objects) and **when the majority of costs are direct materials (DM) and direct labour (DL), manufacturing overhead is combined into a single cost pool** and allocated using **a single allocation base** (or cost driver).
- As complexity increases**, indirect costs are split into pools defined by process stages** (operating departments) and **allocated to cost objects using specific drivers**.

Activity-Based Costing (ABC) Systems

- ABC Systems measure all costs relevant to management and extend cost assignment beyond production, including design, marketing, order processing costs, etc.
- In highly complex situations (many cost objects) and when indirect costs represent a high proportion of total costs, they are first allocated to individual activities within a department, function, plant, etc., and then allocated to cost objects.

Treatment of Indirect Costs in CMS



Indirect Cost Allocation in Traditional CMS

Manufacturing overhead allocation in traditional CMS:

- Allocation using a → **SINGLE BASE**
- Allocation using **MULTIPLE BASES** → use of multiple allocation bases defined with respect to cost pools (indirect cost pools) that can be constructed by process stage → **PROCESS-STAGE ORIENTATION**

Single-Base Allocation

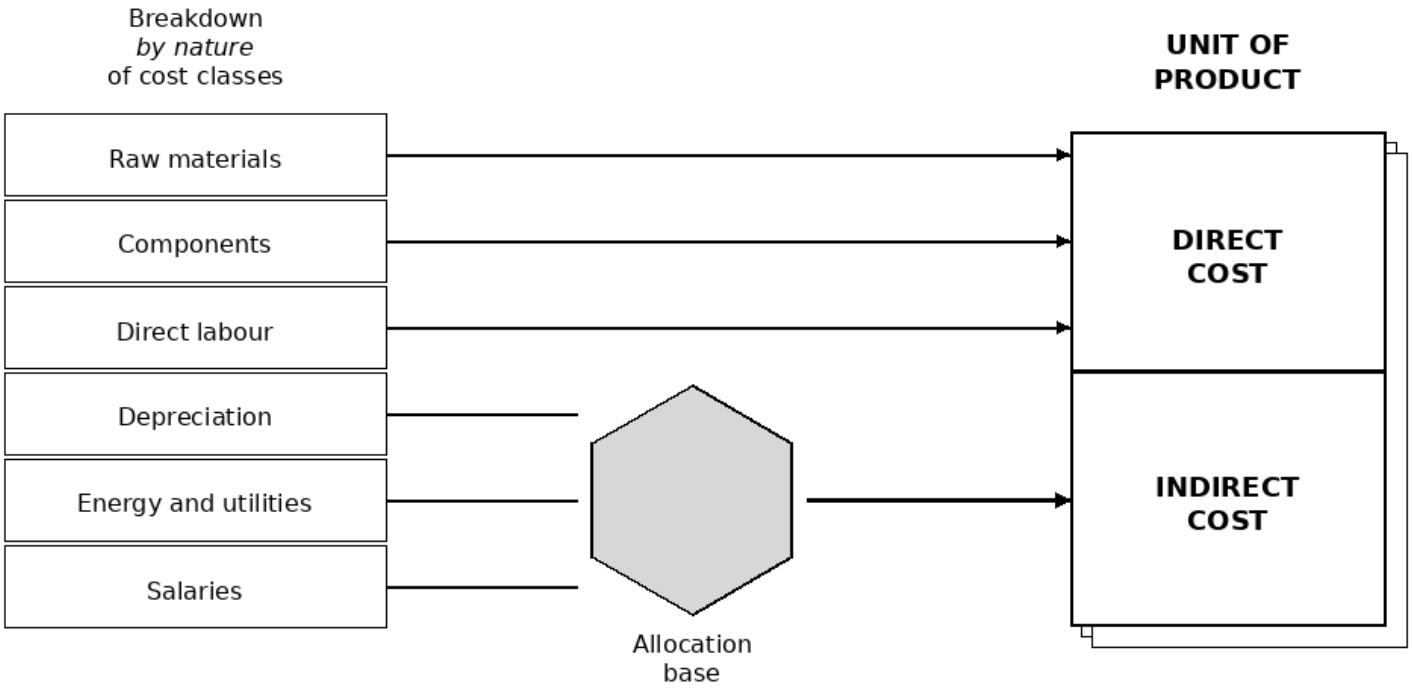
With this method, manufacturing overhead is grouped into a single cost pool, for which a single allocation base is identified (DL hours, DL cost, machine hours, etc.).

Historically, companies tended to use direct labour as the allocation base for indirect costs.

This was because, in labour-intensive production processes, it was reasonable to assume that indirect costs were somehow correlated with labour costs.

Allocation Rate (AR) = $\dfrac{\text{Total manufacturing overhead}}{\text{Total DL hours; Total DL cost; or other allocation base}}$

Single-Base Orientation

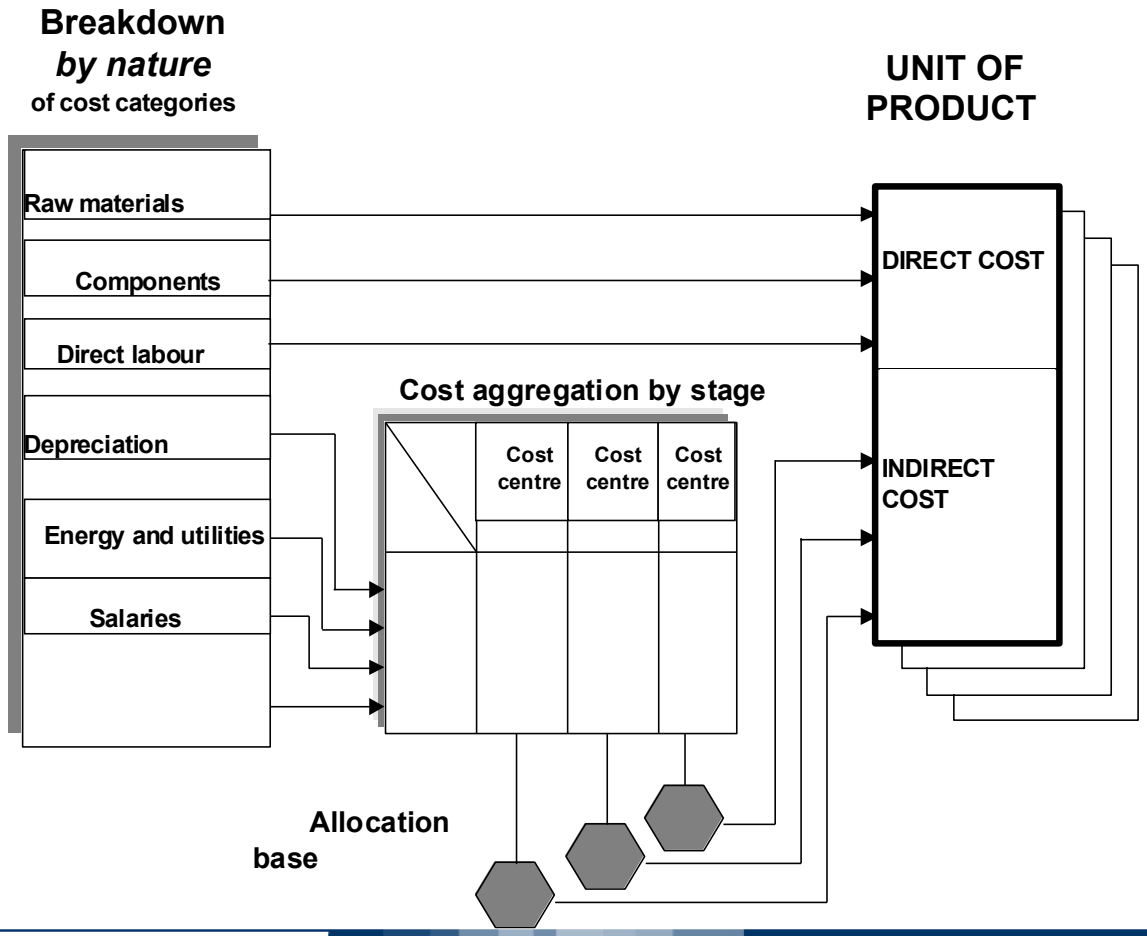


Process-Stage Orientation (Departments)

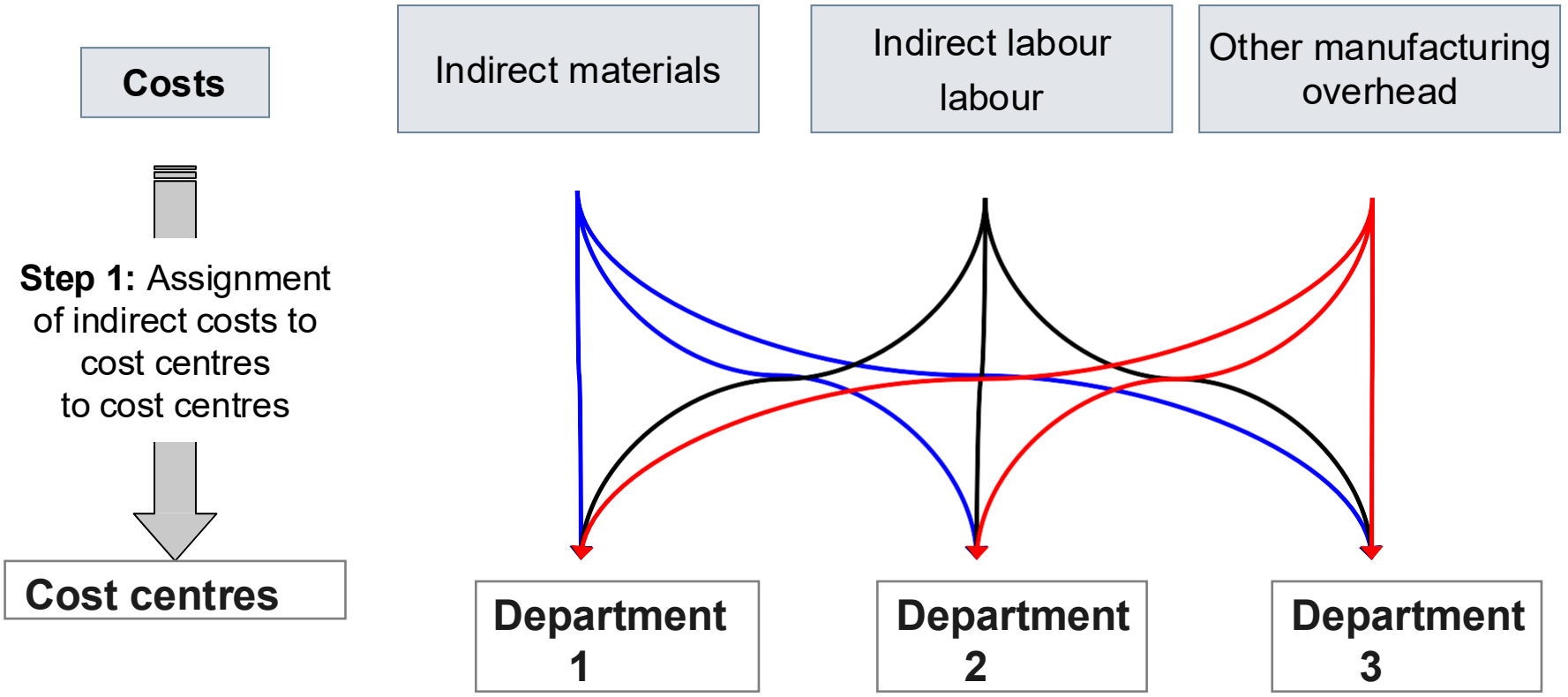
The PROCESS-STAGE orientation is structured in 2 steps:

a) Reclassification of indirect costs by production process stage — this involves aggregating costs by destination, represented by cost centres, which normally correspond to production departments;

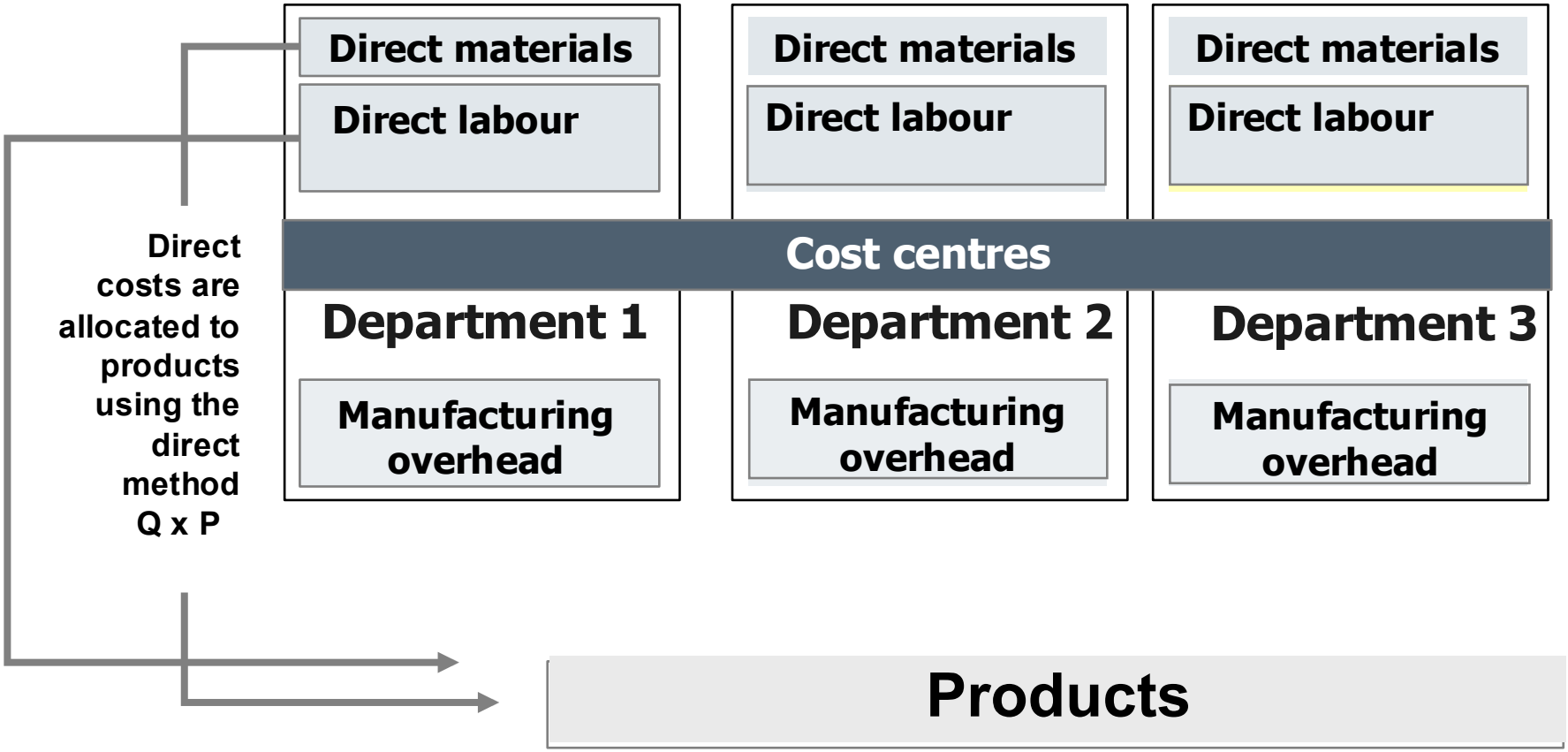
b) Allocation of cost centre pools to the unit of product using allocation rates determined for each centre based on their specific allocation bases



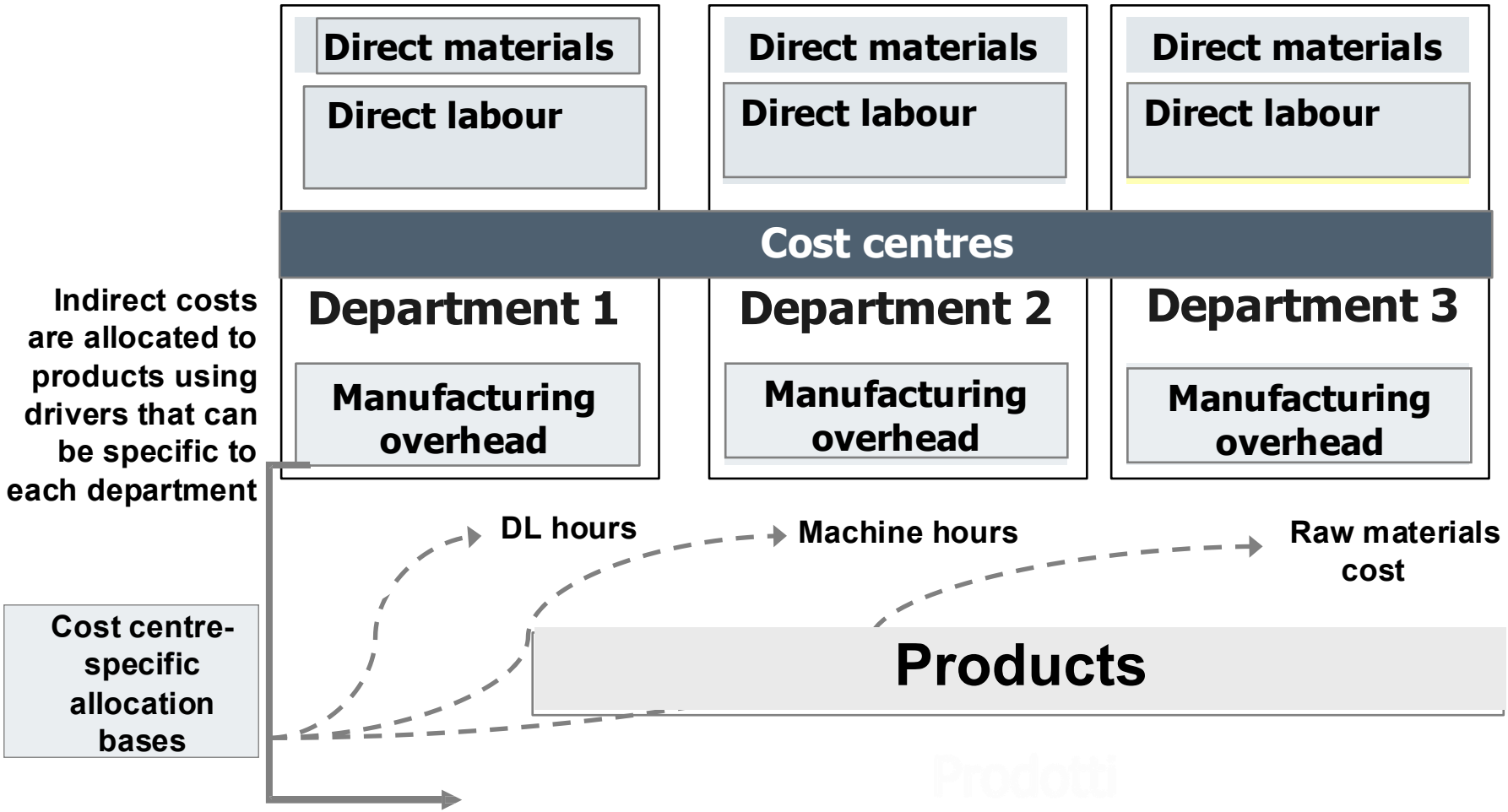
Step 1: Assignment of Indirect Costs to Cost Centres



Step 2: Allocation of Direct Costs to Products



Step 3: Allocation of Department Indirect Costs to Products





CMS Exercise

CMS Exercise – Part 1

Traditional CMS S.p.A. produces 3 products in 2 departments: Machining and Finishing. For a given period, the company employed the following inputs and incurred the following costs:

Direct Costs:		Product			
Machining Department		**P1**	**P2**	**P3**	**TOTAL**
Direct materials:	Quantity (kg)	360	480	360	1200
	Value (€)	€ 14.400	€ 19.200	€ 14.400	€ 48.000
Direct Labour (DL):	Quantity (hrs)	2000	3000	5000	10000
	Value (€)	€ 40.000	€ 60.000	€ 100.000	€ 200.000
Finishing Department		**P1**	**P2**	**P3**	**TOTAL**
Direct materials:	Quantity (kg)	720	360	720	1800
	Value (€)	€ 28.800	€ 14.400	€ 28.800	€ 72.000
Direct Labour (DL):	Quantity (hrs)	625	1125	750	2500
	Value (€)	€ 12.500	€ 22.500	€ 15.000	€ 50.000

CMS Exercise – Part 2

Indirect Costs and their allocation between Departments:		Total Values
Depreciation		€ 100.000
Machining Department	40%	
Finishing Department	60%	
Indirect Labour		€ 80.000
Machining Department	50%	
Finishing Department	50%	
Maintenance		€ 50.000
Machining Department	40%	
Finishing Department	60%	
TOTAL		**€ 230.000**

Single-Base Allocation: Exercise

Using a single plant-wide allocation base (facility / plant)

(facility / plant)

DL cost

Calculation of the Allocation Rate

Single base: DL cost	
Total indirect costs	230,000
Total DL cost	250,000
Coefficient	0.92

Single-Base Allocation: Exercise

To allocate manufacturing overhead to the three products, we need the DL cost for each product.

Indirect Costs:		Product			
Machining department		**P1**	**P2**	**P3**	**TOTAL**
	Quantity (kg)	360	480	360	1200
Direct Materials:	Value (€)	€ 14.400	€ 19.200	€ 14.400	€ 48.000
Direct Labour (DL):	Quantity (hrs)	2000	3000	5000	10000
	Value (€)	€ 40.000	€ 60.000	€ 100.000	€ 200.000
Finishing department		**P1**	**P2**	**P3**	**TOTAL**
	Quantity (kg)	720	360	720	1800
Direct Materials:	Value (€)	€ 28.800	€ 14.400	€ 28.800	€ 72.000
Direct Labour(DL):	Quantity (hrs)	625	1125	750	2500
	Value (€)	€ 12.500	€ 22.500	€ 15.000	€ 50.000

	P1	P2	P3	Total
Direct Costs				
Direct materials	43.200	33.600	43.200	120.000
Direct Labour	52.500	82.500	115.000	250.000
total	**95.700**	**116.100**	**158.200**	**370.000**

Single-Base Allocation: Exercise

Therefore:

Base unica: costo MOD

Costi indiretti totali	230.000
Costo MOD totale	250.000
Coefficiente	0,92

	P1	P2	P3	Total
Direct Costs				
Direct materials	43.200	33.600	43.200	120.000
Direct Labour	52.500	82.500	115.000	250.000
total	***95.700***	***116.100***	***158.200***	***370.000***
Indirect costs of production allocated	**48.300**	**75.900**	**105.800**	***230.000***
Manufacturing total costs	**144.000**	**192.000**	**264.000**	***600.000***

Process-Stage Allocation: Exercise

Total manufacturing overhead of €230,000 is allocated between the two departments:

Indirect cost and their allocation within the departments:		Total Values
Depreciation		€ 100.000
Machining department	40%	
Finishing department	60%	
Indirect Labour		€ 80.000
Machining department	50%	
Finishing department	50%	
Maintenance		€ 50.000
Machining department	40%	
Finishing department	60%	
TOTAL		**€ 230.000**

The controller has noted that the indirect costs of the MACHINING DEPARTMENT are correlated with DL costs, while those of the FINISHING DEPARTMENT are more closely correlated with direct materials costs.

General Indirect costs	Values	Allocation Base		Coeffic. of allocation
Machining department	€ 100.000	DL Costs	€ 200.000	€ 0,500
Depreciation	€ 40.000			
Indirect Labour	€ 40.000			
Maintenance	€ 20.000			
Finishing department	€ 130.000	Direct material Costs	€ 72.000	€ 1,806
Depreciation	€ 60.000			
Indirect Labour	€ 40.000			
Maintenance	€ 30.000			

Process-Stage Allocation: Exercise

Therefore



	Products			
Direct Costs:	**P1**	**P2**	**P3**	**TOTAL**
Direct Materials:	€ 43.200	€ 33.600	€ 43.200	€ 120.000
Direct Labour:	€ 52.500	€ 82.500	€ 115.000	€ 250.000
Total Direct Costs:	**€ 95.700**	**€ 116.100**	**€ 158.200**	**€ 370.000**
Indirect Costs:	**P1**	**P2**	**P3**	**TOTAL**
Machining Department	€ 20.000	€ 30.000	€ 50.000	€ 100.000
Finishing Department	€ 52.000	€ 26.000	€ 52.000	€ 130.000
Total Cost Indirect Allocation	**€ 72.000**	**€ 56.000**	**€ 102.000**	**€ 230.000**
Total Manufacturing Cost	**€ 167.700**	**€ 172.100**	**€ 260.200**	**€ 600.000**

Use of Multiple Allocation Rates for Indirect Costs

- In the presence of a high degree of production activity diversification:
  - **products that differ in their production methods;**
  - production processes with varying degrees of automation;
  - differences in how products use departments/stages;

it may be appropriate to use multiple allocation rates for indirect costs

- The use of multiple ACs* serves a dual purpose:
  - to provide more relevant product/service cost information for cost management and decision support;
  - to facilitate the recording of costs relating to products and services.

Dimensions for structuring indirect cost allocation rates

- Separate ACs by production department
- Separate ACs for variable and fixed indirect costs
- Separate ACs by production plant
- Separate ACs by product line or service class
- Separate ACs for indirect costs linked to materials, labour, or plant

Use of Multiple Allocation Rates for Indirect Costs

- **The allocation of indirect costs accurately reflects the resources (type of departmental DL) and production capacities (departmental technologies) used for each product.**
- The use of different ACs allows the adoption of differentiated allocation bases according to the characteristics of each department (DL hrs – machine hrs – etc.).
- The differentiation of ACs makes it possible to identify the drivers of manufacturing overhead within each department.



Setting separate ACs for each production department is the guiding design principle for effective product costing systems (relevance of cost information):

- it is worthwhile to differentiate ACs by department up to the point at which significant variations in product/service costs are observed,
- the level of differentiation must be evaluated against the trade-off between cost information accuracy and the time/cost of managing the costing system.



Exercise: Full Manufacturing Cost and Process-Stage Indirect Cost Allocation

Exercise

Pinco S.p.A. produces two products (A and B) in two production departments (D1 and D2). The following information is available for the two products:

	PDTA	PDTB	TOT
Single Direct Materials	€ 30,00	€ 10,00	
DL hours D1	4	1	
DL hours D2	1	4	
DL hourly rate			€ 20,00
Production Volume	2.000	5.000	

The company records the following manufacturing overhead:

Manufacturing Overhead	
Depreciation	500.000
Indirect departmental staff	800.000
Maintenance costs	200.000
Department management costs	620.000
Indirect materials costs	80.000
	2.200.000

Exercise

1) Determine the unit full manufacturing cost of both products using a single-base indirect cost allocation method (base: DL hours)
2) Determine the unit full manufacturing cost of both products using a process-stage indirect cost allocation method, assuming that the allocation bases for both production departments are DL hours, and that the departmental indirect costs are as follows:

Indirect costs	D1	D2	Total
Depreciation	200.000	300.000	500.000
Indirect departmental staff	200.000	600.000	800.000
Maintenance costs	80.000	120.000	200.000
Departmental management costs	220.000	400.000	620.000
Indirect materials costs	50.000	30.000	80.000
	750.000	1.450.000	2.200.000

Exercise (Solutions)

1. Product A = € 444,29; Product B = € 424,29

2. Product A = € 426,68; Product B = € 431,33